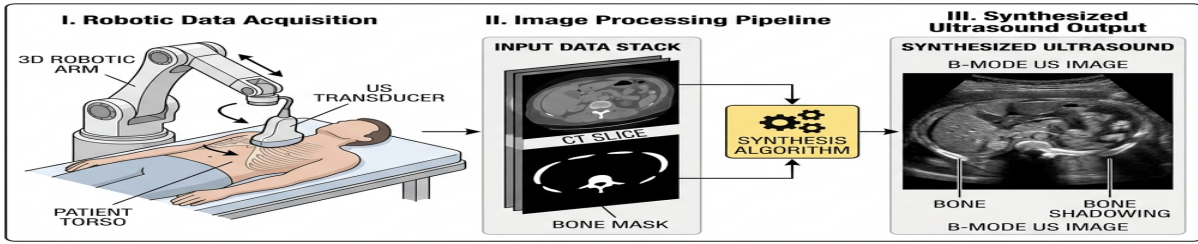


Title: High-Fidelity Real-Time Robotic Ultrasound Simulation via Semantic-Guided Generative Translation

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Robotic ultrasound (US) imaging offers a promising solution for standardized, operator-independent diagnostic scanning and automated medical interventions. However, developing robust control policies requires high-fidelity, computationally efficient simulation environments that can accurately mimic both physical contact dynamics and realistic US image generation. In this work, we present a lightweight, real-time, and motion-adaptive robotic ultrasound scanning simulation framework built in PyBullet, utilizing a clinical curved array transducer probe model mounted on a Franka Panda manipulator. To bridge the gap between simulation and real B-mode US scans, we propose a two-channel semantic-guided generative translation pipeline that combines oblique CT slices with bone segmentation masks. The generative translation network is based on a Pix2Pix architecture trained on the UltraBones dataset. By incorporating intensity histogram matching and highly optimized bilinear interpolation pathways, the framework delivers realistic ultrasound synthesis with sharp acoustic shadowing and specular bone reflections at negligible computational overhead, running entirely on standard CPU hardware. An active compliance controller is integrated to dynamically modulate probe depth based on estimated penetration depth, enforcing clinical safety limits and simulating realistic tissue indentation. Furthermore, we wrap the simulator in a standard OpenAI Gymnasium environment featuring continuous 6-DOF relative controls, multi-modal observation spaces (synthesized B-mode images, contact force, and end-effector pose), and safety-critical terminations. Offline and online evaluations demonstrate that our framework achieves high anatomical consistency and visual fidelity. The reinforcement learning environment achieves a performance benchmark of 3.2 FPS on a single CPU core, while the decoupled GUI visualizer runs at a smooth 60 FPS, establishing an accessible, high-performance platform for training, testing, and validating autonomous robotic scanning policies without the need for local GPU acceleration.